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A/B Testing & Uplift Modeling

E-Commerce Marketing Effectiveness Analysis

April 2026 | Data Science Portfolio Project

588K Users	4 Segments	AUC 0.857	64.5% Persuadable
Dataset Size	K-Means Clusters	XGBoost Score	Uplift Model

Tools: Python • XGBoost • SHAP • Optuna • Scikit-learn • Statsmodels

EXECUTIVE SUMMARY

Statistical significance does not equal business impact.

This project investigates the effectiveness of digital advertising in an e-commerce context using a dataset of 588,101 users exposed to either advertisements or public service announcements (PSA). The analysis spans three layers: classical hypothesis testing, behavioral segmentation, and machine learning-based uplift modeling.

The headline finding is counterintuitive: while the A/B test yields a statistically significant result ($Z = 7.37$, $p < 0.001$), the practical effect size is small (Cohen's $h = 0.053$). The real opportunity is hidden within user segments — in the high-exposure cluster, advertising nearly doubles the conversion rate from 7.8% to 14.4%.

2.55%	h = 0.053	0.857	20–100
Ad Conversion Rate	**Effect Size**	**XGBoost AUC**	**Optimal Ad Band**
vs. 1.79% PSA	Small — Cohen's scale	After Optuna tuning	Ads per user

> ***"Advertising works — but only for the right users, at the right exposure level."***

Key Findings at a Glance

01

A/B Test is significant but effect size is small.

Z-statistic of 7.37 confirms the difference is not due to chance. However, Cohen's h of 0.053 indicates the practical impact is limited when averaged across all users.

02

Ad exposure volume is the strongest conversion driver.

SHAP analysis identifies total_ads as the dominant feature. Users in the high-exposure segment (20–100 ads) show uplift of 13.4%, compared to 0.5% for low-exposure users.

03

Uplift modeling reveals that 35.5% of users are Sleeping Dogs.

Showing ads to these users produces no benefit or negative effect. Excluding them from targeting preserves budget and improves overall campaign ROI.

04

Monday at 16:00 is the optimal targeting window.

Day-hour analysis shows Mondays produce the highest conversion rate (3.28%) and 16:00 is the peak conversion hour (3.08%), suggesting a mid-afternoon, start-of-week effect.

METHODOLOGY

A three-layer analytical framework.

The project is structured as a progressive analytical pipeline. Each layer builds on the previous, moving from descriptive statistics to predictive modeling.

LAYER 1

Hypothesis Testing

Z-test for proportions, Cohen's h effect size, power analysis. Establishes whether observed differences are statistically credible.

LAYER 2

Behavioral Segmentation

K-Means clustering on ad exposure and time features. Reveals heterogeneous treatment effects across user segments.

LAYER 3

ML & Uplift Modeling

XGBoost with SHAP explainability + T-Learner uplift model. Identifies Persuadable users and validates with Qini curve.

Data Overview

The dataset contains 588,101 user records with the following features:

Feature	Type	Range	Description
user id	Integer	Unique	Anonymous user identifier
test group	Categorical	ad / psa	Treatment assignment
converted	Boolean	True / False	Purchase outcome
total ads	Integer	1 – 2,065	Ad impressions shown
most ads day	Categorical	Mon–Sun	Peak exposure day
most ads hour	Integer	0 – 23	Peak exposure hour

Data Cleaning

585 users with more than 500 ad impressions were removed as likely bot or internal test accounts, resulting in a clean dataset of 587,516 records. No missing values were present in the original data.

RESULTS

From statistical significance to business intelligence.

1. A/B Test Results

A two-proportion Z-test was conducted comparing conversion rates between the ad group ($n \approx 400K$) and PSA group ($n \approx 100K$).

Metric	Ad Group	PSA Group	Interpretation
Conversion Rate	2.55%	1.79%	Ad group converts more
Z-Statistic	7.37	—	Highly significant
P-Value	< 0.0001	—	Reject H_0
Cohen's h	0.053	—	Small effect size
Min. Sample Required	5,588	—	Test is overpowered

The test is overpowered: with 588K users, even trivially small differences appear statistically significant. This underscores the necessity of reporting effect size alongside p-values.

2. Segmentation — Heterogeneous Effects

K-Means clustering (k=4) on ad exposure volume and viewing hour reveals that the aggregate A/B result masks dramatically different group-level dynamics.

Segment	Users	Avg. Ads	Ad Conv.	PSA Conv.	Lift
Morning Crowd	272K	15	1.07%	0.71%	+0.36%
Evening Crowd	258K	15	1.43%	1.29%	+0.14%
Super Exposed	7.5K	257	15.46%	14.08%	+1.38%
High Exposed ★	**50K**	**89**	**14.35%**	**7.81%**	**+6.54%**

> ***"The High Exposed segment (Cluster 3) shows a 6.54pp lift — advertising nearly doubles conversion rate in this group."***

3. Machine Learning — XGBoost

An XGBoost classifier was trained to predict individual conversion probability. Hyperparameters were optimized using Optuna (50 trials) with 3-fold cross-validation.

SHAP analysis confirms that total_ads is by far the most predictive feature, followed by most_ads_hour and day_num. The is_ad feature (treatment indicator) ranks last, suggesting that exposure volume — not mere ad exposure — drives conversion.

4. Uplift Modeling — T-Learner

A T-Learner approach trains separate XGBoost models for the treatment (ad) and control (PSA) groups, then computes the incremental lift per user.

User Type	Count	Share	Action
Persuadable	75,837	64.5%	Prioritize for ad targeting
Sleeping Dog	41,667	35.5%	Exclude from campaigns
Avg. Uplift Score	0.049	—	4.9pp lift on targeted users

Qini curve validation shows the model correctly ranks high-uplift users: the top 5,000 users by score achieve a cumulative uplift of 3.5%, significantly above the population average. Uplift decreases as the targeting pool expands, suggesting diminishing returns beyond the top ~30K users.

Target smarter, not broader.

The analysis yields a clear strategic message: blanket advertising to all users is inefficient. The data supports a precision targeting approach centered on three actionable principles.

Recommendation 1: Focus budget on the 20–100 ad exposure band

Users who see 20–100 ads convert at 13.4% uplift vs. 0.5% for low-exposure users. Beyond 100 ads, marginal returns plateau. Concentration in this band maximizes ROI per impression.

Recommendation 2: Deploy uplift model to exclude Sleeping Dogs

35.5% of users show zero or negative response to advertising. Excluding them from campaigns reduces wasted impressions and prevents potential negative brand effects.

Recommendation 3: Time campaigns for Monday afternoons

Conversion peaks on Mondays (3.28%) and at hour 16:00 (3.08%). Scheduling campaigns to maximize impressions in this window can meaningfully lift overall conversion without increasing budget.

Limitations & Future Work

This analysis has three notable constraints. First, the dataset contains no revenue information, making direct ROI calculation impossible. Second, causality between ad volume and conversion cannot be fully established — high-exposure users may self-select. Third, the T-Learner uplift approach is sensitive to model specification; S-Learner or causal forest approaches could be tested as alternatives.

Future extensions include integration of revenue data for full funnel ROI analysis, longitudinal tracking of user behavior post-exposure, and deployment of the uplift model as a real-time targeting API using FastAPI and Supabase.